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MILITARY SPECIFICATION

RESTORER, PULSE FORM, TD-1219/G

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, fabrication and testing necessary for the manufacture of Restorer, Pulse Form, TD-1219/G. This unit is also known as High Speed Pulse Restorer (HSPR) and is one element of the Digital Group Multiplex (DGM) family of cable system equipments developed by the Joint Tactical Command, Control and Communications Agency (JTC^{3}A, formerly TRI-TAC). See 6.1 for intended use of the unit.

2. APPLICABLE DOCUMENT

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issues of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-LC Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N/A FSC 5820

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-C-55583	Cable Assembly, Special Purpose CX-11230( )/G Cable Assembly Adapter CS-10734( )/G
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A Notice 4	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781C	Reliability Tests, Exponential Distribution
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines
MIL-STD-2073-2	Packaging Requirements Code
MIL-STD-45662	Calibration System Requirements

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicition.

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SPECIFICATIONS

Communications-Electronics Command

CR-CX-0173-001	Nuclear Survivability/Vulnerability Provisions for the Family of Digital Group Multiplexer Equipments

DRAWINGS

Communications-Electronics Command

DL-SM-B-875936	Restorer, Pulse Form, TD-1219/G
SM-D-875994	Circuit Card Assembly (CCA), Repeater, High Speed

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2. Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification , however, shall super-sede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The equipment shall be constructed in accordance with DL-SM-B-875936, and the requirements specified herein.

3.2.2 Printed wiring. The printed wiring shall meet the requirements of MIL-STD-454, requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements 30 and 64 of MIL-STD-454.

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3.3 Parts, materials, and processes- general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be furnished in accordance with MIL-F-14072 and drawings DL-SM-B-875936.

3.5 Marking. The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. A unique serial number is required on each nomenclatured item and CCA. The numbers shall be consecutively assigned.

3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each assembled TD-1219 unit shall be electrically preconditioned for a minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as (1) no built-in test equipment (BITE) alarms during burn-in and (2) the successful completion of the operational test of 4.7.6 after burn-in.

3.7 Electrical. (See 4.7)

3.7.1 Electrical current surge and voltage transient protection. (See 4.7.1) The equipment shall meet the requirements of 3.7.6 after being subjected to the test of 4.7.1.1 and 4.7.1.2.

3.7.2 Output voltage level. (See 4.7.2) The output signal voltage shall be from 1.6 to 2.4 volts into 58 ohms.

3.7.3 Output rise and fall times. (See 4.7.3) The output signal rise and fall times shall be from 3 to 8 nanoseconds measured into a 58-ohm resistive termination.

3.7.4 Output signal pulsewidth. (See 4.7.4) The output signal pulsewidth for the positive or negative half-cycle of a conditioned diphase "zero" shall be between 24.7 and 27.3 nanoseconds.

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3.7.5 Output signal droop. (See 4.7.5) The output signal droop shall be less than 10% of the peak-to-peak amplitude for a conditioned diphase "one" measured into 58 ohms.

3.7.6 Operational test. (See 4.7.6) The unit shall operate error-free for 10 seconds with pseudo-random data in a looped-back system.

3.7.7 Receiver sensitivity. (See 4.7.7) The unit shall operate error-free for 10 seconds, in the operational test configuration with a 6dB attenuator inserted 1/4 mile from each regenerator input.

3.7.8 Power supply variation. (See 4.7.8) The unit shall operate error-free for 10 seconds, in the operational test configuration with the dc cable current varied between 85.5 and 94.5 milliamperes.

3.7.9 Fault-detector sensitivity. (See 4.7.9) The unit shall have no output signal with a 40 dB attenuator inserted ahead of either regenerator input in the operational test configuration.

3.7.10 Fault-detector resistance. (See 4.7.9) Voltage drop across a normally operating repeater shall be 10.5V ±10% at nominal current. The dc resistance through each regenerator shall increase by 104 ±2 ohms when conditioned diphase traffic is shut off.

3.7.11 Orderwire connector. (See 4.7.12) A 1000 HZ, 2.0 Vrms signal applied between pins A and D (tied together) and pins E and F (tied together) of the orderwire connector shall appear at each regenerator input and output connector. With the inputs and outputs terminated in 430 ohms, the voltage levels at the regenerator inputs and outputs shall be from 1.5 to 2.0 volts.

3.7.12 Environmental signal detection. (See 4.7.13) The pulse restorer shall be capable of recovering data in a full-duplex circuit from a conditioned diphase signal transmitted over 0.4 km of CX-11230 (see MIL-C-55583) from another pulse restorer (PR) or Cable Driver Modem MD-1024/G or transmitted over 7.6 km of cable with 18 PRs (at the 19th restorer) at 19,200 kb/s ±125 b/s with no errors under the following conditions:

a. With the maximum maintenance orderwire ring tone present in both directions.

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b. With the DGM frame request acknowledge pattern being applied to the equipment side of a MD-1024/G.

c.The received signal level shall be varied +0, -4 dB about the nominal level.

3.7.12.1 Group data error with ring tone present. (See 4.7.12.1) There shall be no errors for a period of 10 seconds at a data rate of 18,720 kb/s.

3.8 Service conditions. (See 4.8) The operational test of 3.7.12 shall be performed before, during, and after each service condition test unless otherwise specified in the appropriate test paragraph.

3.8.1 Temperature. (See 4.8.1)

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as 51.7°C (+125°F) plus the heating effects of solar radiation at a rate of 360 BTU/ft^2/hr. The equipment shall withstand exposure (non-operating) to ambient air temperatures as high as +71.1°C (+160°F).

3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient temperatures as low as -45°C (-50°F). The equipment shall withstand exposure (non-operating) to ambient air temperatures as low as -57°C (-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.4 Dust. (See 4.8.4) The equipment shall with-stand, in both operating and non-operating conditions, exposure to fine dust particles with wind speeds of 3542 ft/min (35 knots).

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3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of a salt fog atmosphere.

3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.8.6. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.8.7 Immersion. (See 4.8.7) Prior to and following vibration and shock testing, the equipment in its transit case shall be immersion-proof to a covering depth of 3 feet of water, for a period of not less than 2 hours.

3.8.8 Vibration and shock. (See 4.8.8) The equipment shall withstand vibration and shock induced during field transport by military vehicle and by rough handling. The equipment shall withstand vibrations encountered during transportation and shocks encountered during servicing.

3.9 Electromagnetic interference (EMI). (See 4.9) The unit shall meet the following emission and susceptibility requirements of MIL-STD-461A, Notice 4:

RE02	(14 kHz to 1 GHz)
RE02.1	(14 kHz to 1 GHz)
RS02	
RS03	(10 kHz to 1 GHz)

3.9.1 Bonds and grounds. (See 4.9.1) Bonding and grounding impedance shall not exceed 2.5 milliohms.

3.10 Reliability. (See 4.10) The unit shall have a specified Mean-Time-Between-Failure (MTBF), θ_o of 50,000 hours when in the normal modes of operation.

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies, and replaceable parts shall meet requirement 7 of MIL-STD-454.

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3.12 Unit subassembly electrical tests. (See 4.12) The unit subassembly listed below shall meet the requirements of its test requirement specification (TRS) drawing, as referenced in the following subassembly drawing:

CCA	DRAWING
Repeater, High Speed	SM-D-875994

3.13 Operational tests. Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.6 after all other inspections and tests are completed to determine that the unit is operating prior to packaging.

3.14 System safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirements 1 and 8, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g. luminous dials/marking electron tubes/spark gaps, lenses).

3.15 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with the applicable portions of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires, electrical

c. Wire (harnessing)

d. Plastic material and parts

e. Rubber

f. Wiring and cabling

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g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

3.16 Nuclear survivability. The equipment shall satisfy the Electromagnetic Pulse (EMP) and Initial Radiation requirements stated in Section 3 of CR-CX-0173-001.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging. (See 4.5)

(2) Packaging inspection. (See 4.14)

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspections shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of 46 first article units of the complete equipment. The first article quantity shall be as specified in the contract or purchase order.

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4.3.2 First article inspection. The first article inspections shall consist of inspections specified in table I and shall be performed in the order specified therein except that all hardwired and plug-in assemblies shall undergo in process Group A inspection prior to burn-in.

4.3.3 First article inspection data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

TABLE I. First article inspection.

Inspection	Reqt Para	Test Para	Unit 1	2	3	4	5	6	7	8-46	47+
1. Inspections by Subsidiary Documents	N/A	4.4	Inspections to be performed on all units.								
2. Burn-In	3.6	4.6	Inspections to be performed on all units.								
3. Group A Inspection	See Table II		Inspections to be performed on all units.								
4. Group B Inspection	See Table III		Inspections to be performed on all units.								
5. Group C Inspection											
Subgroup 1											
High Temperature	3.8.1.1	4.8.1.1	1								
Low Temperature	3.8.1.2	4.8.1.2	2								
Subgroup 2											
EMI	3.9	4.9		1							
Immersion	3.8.7	4.8.7			1						
Vibration/Shock	3.8.8	4.8.8			2						
Immersion	3.8.7	4.8.7			3						
Subgroup 3											
Humidity (Note C)	3.8.2	4.8.2				1					

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TABLE I. First article inspection (continued)

Inspection	Reqt Para	Test Para	Unit 1	2	3	4	5	6	7	8-46	47+
Subgroup 4											
Altitude	3.8.3	4.8.3					1				
Dust (Note C)	3.8.4	4.8.4					2				
Salt fog (Note C)	3.8.5	4.8.5					3				
Fungus (Note C)	3.8.6	4.8.6						1			
Subgroup 5											
Electrical Current Surge and Voltage Transient Protection	3.7.1	4.7.1							1		
Receiver Sensitivity	3.7.7	4.7.7							2		
Power Supply Variation	3.7.8	4.7.8							3		
Fault Detector Sensitivity	3.7.9	4.7.9							4		
Fault Detector Resistance	3.7.10	4.7.10							5		
Orderwire Connector	3.7.11	4.7.11							6		
Dimensional Interchangeability	3.11	4.11							7		
Output Signal Rise and Fall Time	3.7.3	4.7.3							8		
Output Signal Pulse Width	3.7.4	4.7.4							9		
Output Signal Droop	3.7.5	4.7.5							10		
6. Group D Inspection											
Reliability	3.10	4.10									39 units

NOTES:

a. The inspections 1 through 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

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b. The number in the unit columns in the table covered by Inspections 5 and 6 specify the order of inspections for the indicated unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. All first article units shall be subjected to Group A and Group B inspection.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

	Where required
Printed wiring.	3.2.2
Semiconductor devices. and integrated circuits	3.2.3
Parts, materials, and processes	3.3
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in Table II shall be inspected for conformance to the inspections specified in Table II.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in table II. The order of inspection of the subassemblies, assemblies, and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table II shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in table II.

b. Only for spares which are not installed into complete units, do the following: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in table II (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies and subassemblies which have passed the 100% visual/mechanical and electrical inspections of step a above shall be assembled into the end equipments, which then shall be subjected by the contractor to the burn-in test requirements of 4.6 and an end equipment 100% visual/mechanical and electrical inspection in accordance with table II, in that order.

d. After completion of the above 100% inspections by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in table II. Group A electrical inspection shall be performed in the order shown in table II.

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TABEL II. Group A inspection.

Inspection	Reqt Para	Insp Para	AQL Major	AQL Minor
VISUAL-MECHANICAL (Including Spare Assemblies)				
Repeater CCA (2 Identical)	3.15	4.12	2.5%	4%
ELECTRICAL PERFORMANCE (Including Spare Assemblies)				
Repeater CCA (2 Identical)	3.12	4.13	1.0%	*
ELECTRICAL				
Operational Tests	3.7.6	4.7.6	1.0%	*

*All electrical defects are Major.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in table III and the inspection level shall be S-4.

4.5.5.2 Order of inspection within Group B. Group B inspection shall be performed in the order listed in table III.

TABLE III. Group B inspections.

Inspection	Reqt Para	Insp Para	AQL
Output Voltage Level	3.7.2	4.7.2	4.0%

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4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspections.

4.5.3.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature testing in that order.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month, and one unit shall be selected every eight month. The unit selected every fourth month shall be subjected to immersion, vibration, shock and immersion testing with the unit subjected to testing in that order. The unit selected every eighth month shall be subjected to EMI testing.

4.5.3.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity testing.

4.5.3.1.4 Subgroup 4. Two units shall be selected from the first production lot for this subgroup. Two units shall be selected every twelfth month for subsequent Group C inspections. One unit shall be subjected to Altitude, Dust and Salt Fog testing and one unit shall be subjected to fungus testing.

4.5.3.1.5 Subgroup 5. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month. The unit shall be subjected to the electrical tests in the order shown in table IV.

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TABLE IV. Group C inspection.

Inspection	Reqt Para	Insp Para
Subgroup 1		
High Temperature	3.8.1.1	4.8.1.1
Low Temperature	3.8.1.2	4.8.1.2
Subgroup 2.		
EMI	3.9	4.9
Immersion	3.8.7	4.8.7
Vibration	3.8.8.1	4.8.8.1
Shock	3.8.8.2	4.8.8.2
Immersion	3.8.7	4.8.7
Subgroup 3.		
Humidity	3.8.2	4.8.2 (Note 1)
Subgroup 4.		
Altitude	3.8.3	4.8.3
Dust	3.8.4	4.8.4 (Note 1)
Salt Fog	3.8.5	4.8.5 (Note 1)
Fungus	3.8.6	4.8.6 (Note 1)
Subgroup 5.		
Electrical Current Surge and Voltage Transient Protection	3.7.1	4.7.1
Receiver Sensitivity	3.7.7	4.7.7
Power Supply Variation	3.7.8	4.7.8
Fault Detector Sensitivity	3.7.9	4.7.9
Fault Detector Resistance	3.7.10	4.7.10
Orderwire Connector	3.7.12	4.7.12
Dimensional Interchangeability	3.11	4.11
Output Signal Rise and Fall Time	3.7.3	4.7.3
Output Signal Pulse Width	3.7.4	4.7.4
Output Signal Drop	3.7.5	4.7.5

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NOTE 1. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed for each subgroup in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, samples units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is reworked to specifications and the sample units are resubjected to and passed Group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the test specified in table V and shall be performed on units from lots which have been subjected to and have met Group A and Group B inspection. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE V. Group D inspections.

Inspection	Reqt Para	Insp Para
Reliability	3.10	4.10

4.5.5.1 Sampling for Group D inspection. Thirty-nine units shall be selected from the first month's production and subjected to the test of 4.10. For subsequent Group D inspections, a maximum of 39 units shall be selected at random from every twelfth month's production.

4.5.5.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.5.5.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is reworked to specifications and the sample units are resubjected to and pass Group A inspection only.

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4.6 Electrical preconditioning (burn-in). (See 3.6) During electrical preconditioning, the equipment shall be connected as per figure 1. Test conditions of 4.10.3 shall apply except that duty cycle shall be 100. If a failure occurs during this preconditioning, the failure shall be recorded, the failed components replaced, and the preconditioning cycle reinstituted. Failure data obtained shall be submitted with the lot containing the unit in which the failure occurred. The requirements of 3.6 shall be met.

4.7 Electrical. (See 3.7) Unless otherwise specified all tests shall be performed at 115V ±5% from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipment listed in table VI or their equivalent will be utilized during testing and will be calibrated in accordance with MIL-STD-45662A. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. Unless otherwise specified, special test equipment (see table VII) shall be provided by the contractor subject to approval by the Government. The Government shall be responsible for supplying the Government Furnished Equipment (GFE) required for test listed in table VIII. Table IX identifies the external interface connections of the equipment as referenced in the test configuration figures. Unless otherwise noted, all resistors shown in the figures are 0.5 watts.

TABLE VI. Standard test equipment.

Description	Mfg/Model No. (or equivalent)
Volt-Ohm Meter	Simpson 260
DC Milliameter/Voltmeter	H.P. 412A
Audio Oscillator	H.P. 200CD
Attenuator	H.P. 4436A
AC Voltmeter	H.P. 403
Oscilloscope	Tektronix 7904
DC Power Supply	Kepco DW-91

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TABLE VII. Special test equipment.

Equipment Name	Drawing No.	Functional Description
Group Simulator	A-3051667	The Group Simulator simulates the group data, orderwire and telemetry signals. It operates either a long or short pseudo-random sequence. The three signals can be generated simultaneously or in any combination. A TTL error output signal is provided for counting errors externally.

TABLE VIII. Government furnished equipment (GFE)

Equipment	Drawing No.	Functional Description
Cable Driver Modem MD-1024/G	SM-D-875933	The MD-1024/G is used to generate line-type signals so that the equipment side interface of the UUT can be tested.
CX-11230/G 1/4 mile reel		Twin coax cable supplied as GFE for test.

TABLE IX TD-1219 interface connections.

Connector	Function	Hi	Ret	Gnd	Signal Type
A	Group Data In	*(M)			Diphase
	Group Data Out	*(F)			Diphase
B	Group Data In	*(M)			Diphase
	Group Data Out	*(F)			Diphase
	A-B Current Monitor	A	B	F,E	
Orderwire	A-B Output Monitor	G			
	B-A Current Monitor	D	C	F,E	
	B-A Output Monitor	H			

*(M) Male, (F) Female; Type determined by center conductor.

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4.7.1 Electrical current surge and voltage transient test. (See 3.7.1)

4.7.1.1 Overcurrent protection.

a. Connect a DC power supply through 1/4 mile of CX-11230 cable to the UUT as shown in Figure 2.

b. With the switch open, adjust the power supply so that the voltage at the output of the UUT is 500 V.

c. Close and open the switch five times at approximately 1-second intervals.

d. Set the CABLE CUR switch to OFF.

e. Reverse the CABLE connections to the UUT and repeat Steps a through d.

f. Verify that the UUT is undamaged after steps c and e.

4.7.1.2 Overvoltage protection.

a. With the shorting switch open, set the power supply voltage to 1000V for at least 5 seconds.

b. Turn OFF the DC Power Supply.

c. Reverse the CABLE connections to the UUT and repeat Steps a through c.

d. Verify that the UUT is undamaged after a. and c.

4.7.2 Output voltage level. (See 3.7.2)

a. Connect the equipment as shown in figure 3. Set the MD-1024/G rate switch to position 14 (4608 kb/s).

b. Set the CABLE CUR switch to ON.

c. Using the oscilloscope, measure the peak-to-peak voltage level across the 58-ohm termination. See figure 4.

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d. Verify that the specified requirements are met.

e. Set the CABLE CUR switch to OFF.

f. Reverse the CABLE connections to the UUT and repeat Steps b. through e.

4.7.3 Output rise and fall times. (See 3.7.3)

a. Connect the equipment as shown in figure 3. The switch positions are the same in 4.7.2.

b. Set CABLE CUR switch to ON.

c. Using the oscilloscope, measure the rise and fall times between the 10% and 90% points of the slopes of the loaded output signal level of the UUT.

d. Verify that the rise and fall times meet the specified requirements.

e. Set the CABLE CUR switch to OFF.

f. Reverse the CABLE connections to the UUT and repeat Steps 2 through 5.

4.7.4 Output signal pulsewidth. (See 3.7.4)

a. The equipment connections, and switch positions are the same as in 4.7.2.

b. Set CABLE CUR switch to ON.

c. Measure the pulsewidths of the positive and negative half cycles at a diphase "zero". (See figure 4)

d. Verify that the pulsewidth meet the specified requirements.

e. Set the CABLE CUR switch to OFF.

f. Reverse the CABLE connections to the UUT and repeat Steps b. through d.

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4.7.5 Output signal droop. (See 3.7.5)

a. The equipment connections, and switch positions are the same as in 4.7.2.

b. Set CABLE CUR switch to ON.

c. Measure the droop as shown in figure 4 for a conditioned diphase "one".

d. Verify that the droop meets the requirements.

e. Set the CABLE CUR switch to OFF.

f. Reverse the CABLE connections to the UUT and repeat Steps b. through e.

4.7.6 Operational test. (See 3.7.6)

a. Connect the equipment as shown in figure 5.

b. Set the Group Simulator and the MD-1024/G to 18,720 kb/s.

c. Turn on the Power Switch and the Cable Current Switch on the MD-1024/G. Monitor the group error rate for a period of 10 seconds. Verify that there are no errors.

d. Using the Fault Locate circuitry on both MD-1024/G's, verify that there are no faulty restorers.

FAULT = N + 1
N = Number of Restorers

4.7.7 Receiver sensitivity. (See 3.7.7) Verify that 4.7.1 was performed prior to conducting this test.

a. Connect the equipment as shown in figure 6. Set the MD-1024/G rate switch to position 14 (4608 kb/s).

b. Insert the 6-dB attenuator in series with the input cable to the UUT.

c. Set the CABLE CUR switch to ON.

d. Verify that there are no errors in 10 sec, as required.

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e. Set the CABLE CUR switch to OFF.

f. Reverse the CABLE connections on the UUT, and repeat steps b through d. Verify that there are no errors in a 10 second period.

4.7.8 Power supply variation. (See 3.7.9)

a. Connect the equipments as shown in figure 6. set the MD-1024/G rate switch to position 14 (4608 kb/s).

b. Insert the cable current attenuator in series with the input of UUT.

c. Set the CABLE CUR switch to ON.

d. Adjust the cable current measured on the milliammeter to 85.1 mA.

e. Verify that there are no errors over a 10 second period.

f. Set the CABLE CUR switch to OFF.

4.7.9 Fault-detector sensitivity. (See 3.7.9)

a. Connect the equipment as shown in figure 7. Set the MD-1024/G rate switch to position 14 (4608 kb/s).

b. Insert the 40-dB attenuator in series with the input to the UUT.

c.Set the CABLE CUR switch to ON.

d. Verify that the LS Alarm (Loss-of-Signal) is activated on the MD-1024/G.

e. Set the CABLE CUR switch to OFF.

4.7.10 Fault-detector resistance. (See 3.7.10)

a. Connect the equipment as shown in figure 8. Set the MD-1024/G rate switch to position 14 (4608 kb/s).

b. Set the CABLE CUR switch to ON.

c. Measure the cable current and the voltage at the input of the UUT.

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d. Calculate the resistance of the UUT.

e. Remove Traffic. (Use the CDM Fault Locate Circuitry).

f. Repeat steps c and d..

g. Subtract the resistance calculated in step e from the resistance calculated in step f and verify that this change in resistance is within the required limits.

h. Set the CABLE CUR switch to OFF.

i. Reverse the CABLE connections to the UUT, and repeat steps b through h.

j. This test will be conducted one time only during first article. Measure the voltage drop across the repeater. Verify that the voltage drop is 10.5 V ±10%.

4.7.11 Orderwire connector. (See 3.7.11)

a. Connect the equipment as shown in figure 8.

b. Apply 1000 Hz, 2.0 ±0.1 Vrms between pins (A-D) (tied together) and pins (E-F) (tied together) of the pulse restorer ORDERWIRE connection.

c. Terminate the input and output cable connections in 430-ohms.

d. Measure the voltage across each of the four 430-ohm terminations.

e. Verify the levels meet the requirements of 3.7.11.

4.7.12 Environmental signal detection. (See 3.7.12)

a. Connect the equipment as shown in figure 9.

b. The MD-1024 switch positions are the same as in 4.7.2, except that the RATE SELECT switch on the TRF XMT CCA shall be set to position 17 (18.72 Mb/s).

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c. Connect LINE SIG OUT and the LINE SIG IN on the MD-1024 through the HVBNC cable to the TD-1219 UUT via a 1/4 mile length of CX-11230 cable.

d. Loop back connector B on the UUT via 1/4 mile of CX-11230 cable.

e. Connect the GROUP SIMULATOR to the MD-1024/G and set the GROUP RATE to 18.72 Mb/s.

f. Set the audio oscillator to -4 dBm at 1600 Hz while conected to the AVOW input.

g. Place the CABLE CUR switch on the MD-1024 to the ON position.

h. Monitor the group data for a period of 10 seconds. Verify that no errors are generated.

i. Repeat step h with 4 dB attenuation inserted in the cable path from the MD-1024 to the A side of the UUT.

j. Reverse the connections to the UUT and repeat step h.

4.8 Service conditions. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2, Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +51.7°C (+125°). Maintain internal chamber temperature for six hours at +51.7°C (+125°F). Raise the internal chamber temperature to +82.2°C (+180°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +51.7°C (+125°) within a time period of one hour. Repeat the above steps two additional times to make a total of three 12-hour cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +82.2°C (+180°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

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4.8.1.2 Low temperature. (See 3.7.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -57°C (-70°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure I, perform Procedure II - Operation (paragraph II-3.2 of Method 502.2). Chamber temperature in step 1 of Procedure II shall be adjusted to -45°C (-50°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III. After completion of Procedure III, the test item shall be subjected to Procedure II. In step 4, cycle 5 of table 507.2-I shall be used.

4.8.3 Altitude. The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,500 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 523 mm of Hg (2016 inches of Hg or 10,000 feet above sea level) at a rate not to exceed 2,500 fpm.

4.8.4 Dust. (See 3.8.3) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. The air velocity in step 2 shall be 3542 $\pm$ 250 feet per minute. In step 5, reduce the air velocity to 300 $\pm$ 200 feet per minute and adjust the air temperature to + 63°C(+145°F). The equipment shall be reoriented periodically to equally expose all sides of the equipment to the dust stream. Equipment shall be operated before step 1, during step 8 and during step 11. There shall be no degradation in performance as a result of the dust test, and there shall be no binding or blocking of moving parts, non-operation of contacts or relays, clogging or dust accumulation which could form the nucleus for the collection of water vapor as a result of dust penetration of the test item.

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4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48 hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water not warmer than +38°C (+100°F) and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall also be no clogging or binding of moving mechanical parts.

4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery-condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies with mycelial development in non-critical areas. Critical areas include: terminal spacing, CCAs etc. The equipment shall fail if it shows more than the growth specified above.

4.8.7 Immersion. (See 3.8.7) The equipment shall be subjected to the test of MIL-STD-810D, Method 512.2, Procedure I.

4.8.8 Vibration and shock. (See 3.8.8)

4.8.8.1 Vibration. The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3 for Category 1 -Basic Transportation. The equipment in its transit case, if any, as restrained and/or tied down in its combat transport configuration shall be subjected to the random vibration environment for mission/field transportation. The vibration environment shall be simulated using the test schedules given in figures 514.3-4, 514.3-5, 514.3-6, and figures 514.3-10 through 514.3-22. Test duration for figures 514.3-4, 514.3-5, and 514.3-6 shall be 48 minutes per axis, for a test time of 144 minutes. Test duration for figures 514.3-10 through 514.3-22 shall be as given in the test schedules, for a test time of 3 hours. Total test time shall be 324 minutes. The test item shall not be operated during this test. The equipment in its transit case, if any, shall also be subjected to the test of MIL-STD-810D, Method 514.3 for transportation vibration, Category 3 -Loose Cargo Transport.

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Prior to and following the tests for both Category 1 and Category 3, the equipment shall be operated in accordance with MIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance. The equipment, including transit case, shall not sustain any mechanical damage.

4.8.8.2 Drop. The unit shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure IV, except that the pulse restorer, with end caps in place, shall be subjected to six drops from a height of 20 feet. Two drops will be made on each end of the pulse restorer. Two drops will be on the side of the unit, one on the plane of the CCA, and one perpendicular to the plane of the CCA. The equipment shall not have any degradation in performance, and shall not sustain any mechanical damage.

4.8.8.3 Bench handling. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI. Prior to and following this test, the equipment shall be operated in accordance with MIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance and shall not sustain any mechanical damage.

4.9 EMI tests. (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, and a Government approved test plan.

4.9.1 Bonds and grounds. (See 3.9.1) Prior to performance of EMI testing, compliance with requirements of 3.9.1 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.9.2 General test conditions. The TD-1219 shall be tested for compliance without shielding or other suppression devices on the GFE cables and connectors.

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4.10 Reliability testing. (See 3.10)

4.10.1 The first article reliability test. The first article reliability testing shall be in accordance with Test Plan VIC, of MIL-STD-781C.

4.10.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan VIC, of MIL-STD-781C.

4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following duty cycle: Apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF for 2 hours. Repeat this sequence for each 24-hour cycle. Chamber temperature shall be 50°C to 55°C (122°F to 131°F). Vibration shall be 2.2 G $\pm$ 10% per acceleration value at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.

4.10.4 Reliability test method. The detailed test procedures, and conditions for performing the tests outlined in 4.10.1 and 4.10.2 shall be in accordance with a contractor prepared, Government-approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.6.

4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781C. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

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4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of connectors, fasteners, and mountings which receive mating parts or assemblies.

4.12 Electrical subassembly tests. (See 3.12) The unit subassemblies, including spares, shall be test in accordance with their respective TRS drawings listed in 3.12.

4.13 Visual and mechanical. (See 3.14 and 3.15) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (System Safety Engineering) which can be determined visually.

4.13.1 System Safety Design Verification Check List. An inspection shall be performed, on one unit only, during first article testing, utilizing SEL Form 1183, to verify compliance with 3.14 (System safety engineering.)

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.14.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

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4.14.2 Quality conformance inspection of packaging.

4.14.2.1 Materials inspection. All material to be used in packaging shall be inspected in accordance with the applicable material specification.

4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with Group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examination specified in table X, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at on time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 % defective.

TABLE X Packing inspection provisions.

NO.	Characteristic	Method of Inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight and Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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4.15 Serial numbers. The equipment shall be inspected to verify that the serial numbers appear on each nomenclatured item, except cords and cables.

4.16 Nuclear test. Applicable tests for nuclear survivability shall be performed on the equipment in accordance with Section 3 of CR-CX-0173-001.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2 tables I through VII. coded as follows:

5.1.1 Level A. 10-1-00-OO-NS-X-ED.

5.1.2 Level B. Same as level A.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The TD-1219/G is a field exposed pulse restorer for use in JTC^{3}A repeatered coaxial cable systems. Used with CX-11230 coax cable and the MD-1024/G, it carries digital traffic at 19.2 Mb/s and an analog orderwire. Provides pulse regeneration at 1/4 mile intervals for digital traffic and line loading for orderwire signals.

6.2 Ordering data. Procurement documents should specifiy the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see Section 5).

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c. When first article rough handling tests are not required.

d. Quantity of first article units required.

e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirements.

i. MIL-Q-9858.

j. Length of cables (if any are required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 *Environmental.* Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Final inspection. (See 4.5.3.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. (See 4.5.3.3) Initial shipments of end items will be contingent upon successful completion of all first artice tests.

6.5 Location of operational inspection. (See 4.7.6) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. the amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in table VIII.

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Subject term (key word) listing.

Conditioned Diphase
Digital Group Multiplex
Digital Traffic
High Speed Pulse Restorer (HSPR)
Orderwire
Regeneration
Restorer
Restorer, Pulse Form
Restorer, Pulse Form, TD-1219/G

CUSTODIAN:
ARMY -CR

Preparing Activity
ARMY-CR
Project Nr. 5820-A828

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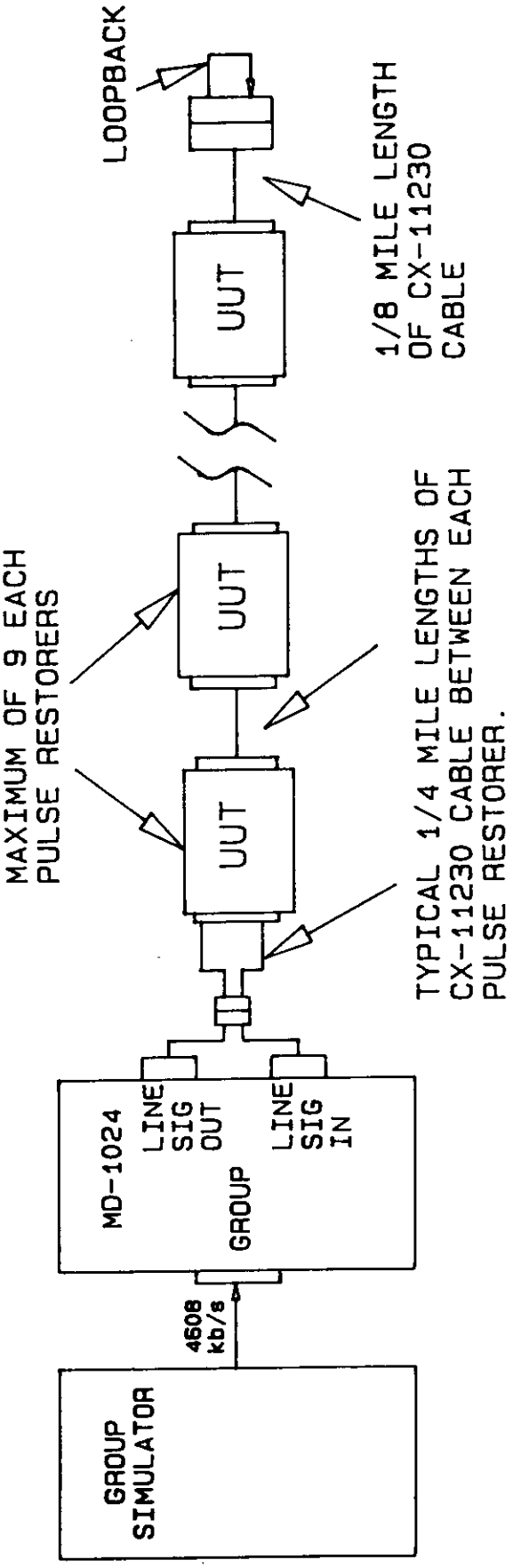
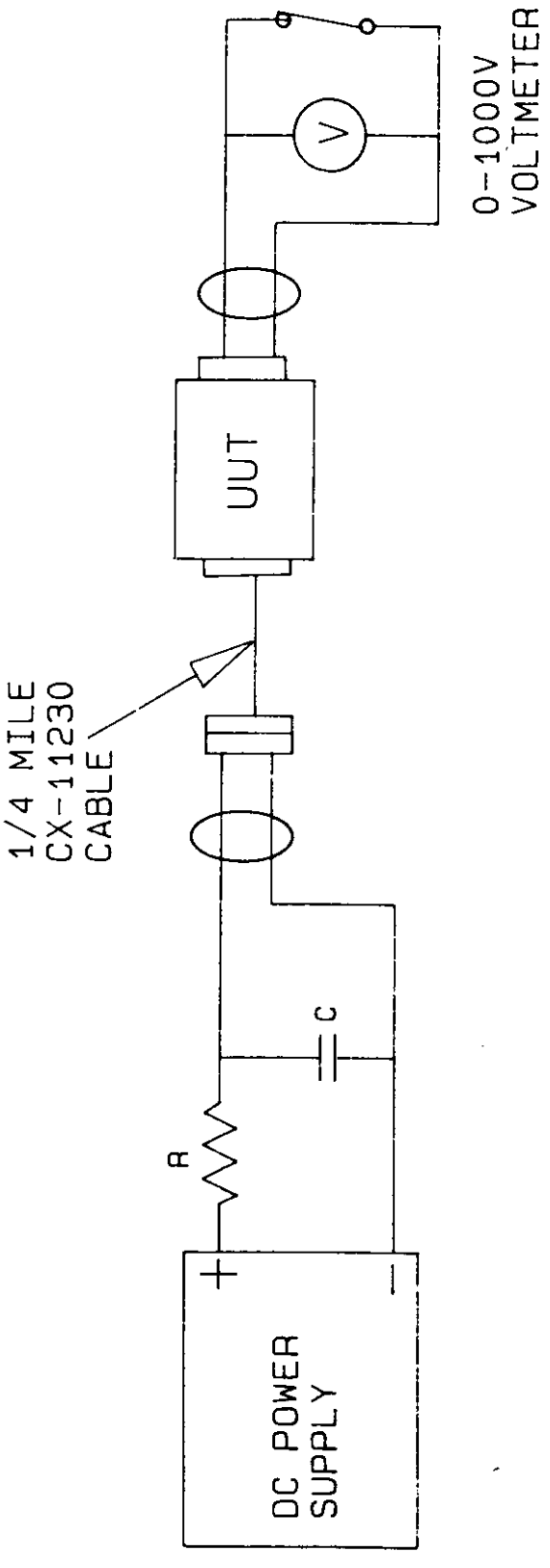


FIGURE 1. Burn-in.

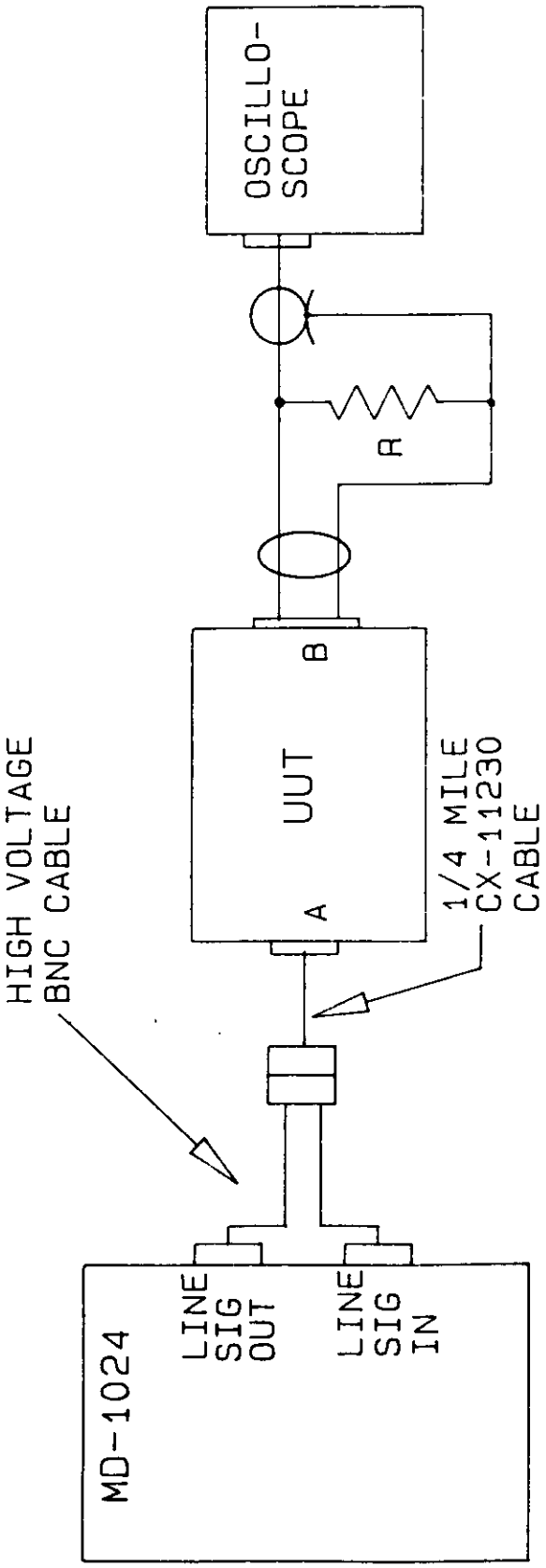
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R 50K OHMS ±1% .50W
C 1.0 uF, 1500V

FIGURE 2. Overcurrent protection; overvoltage protection.

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R 58 OHMS ± 1%

WARNING

AN OPEN CIRCUIT POTENTIAL OF 1000 VOLTS APPEARS ON THE CABLE AND ON LINE SIG OUT CONNECTOR WHEN CABLE CURRENT SWITCH IS ON.

FIGURE 3. Output voltage level; rise and fall time.

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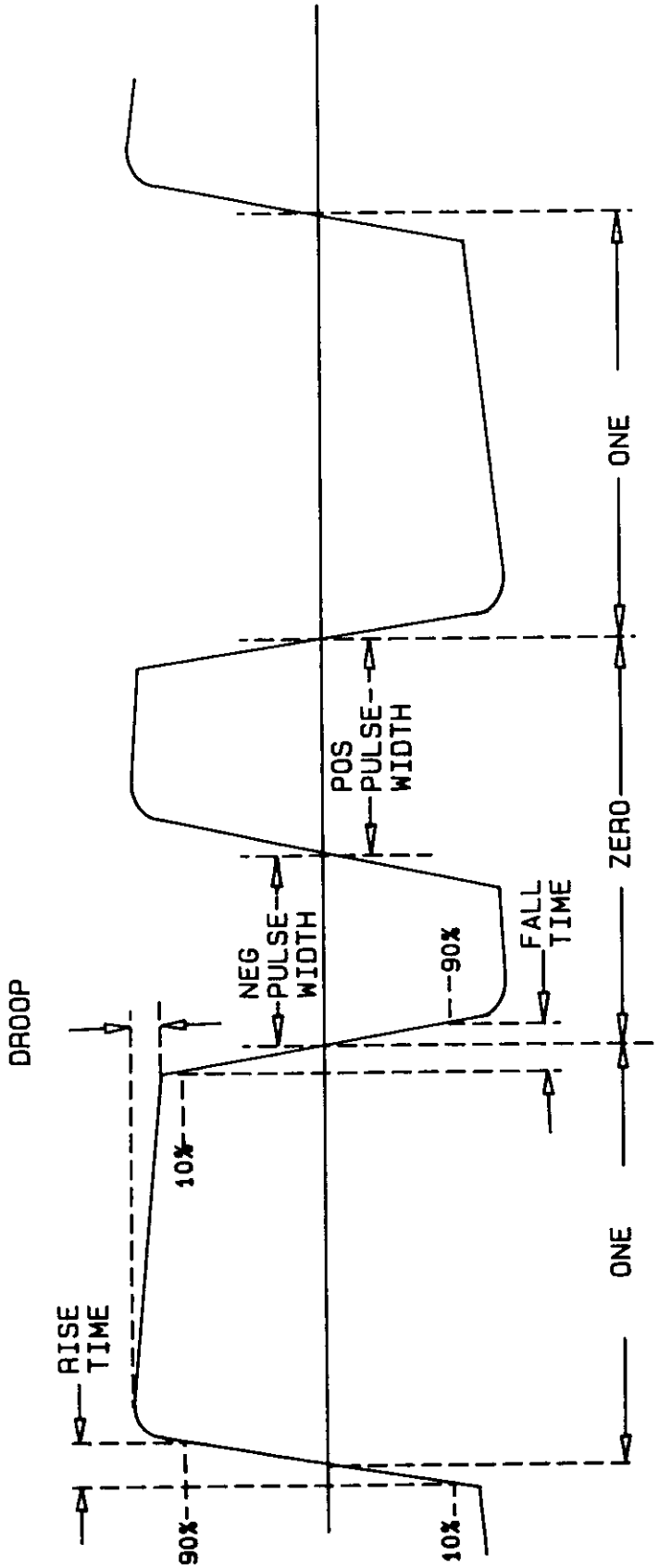


FIGURE 4. Diphase waveform (digital zero).

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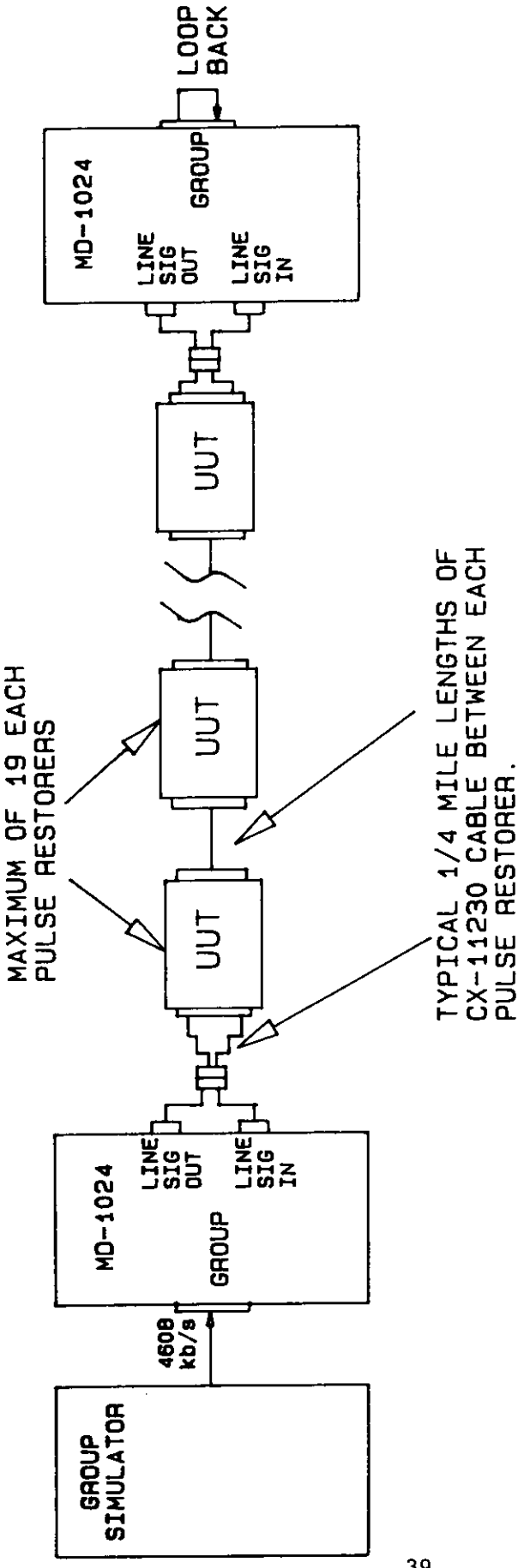


FIGURE 5. Operational test.

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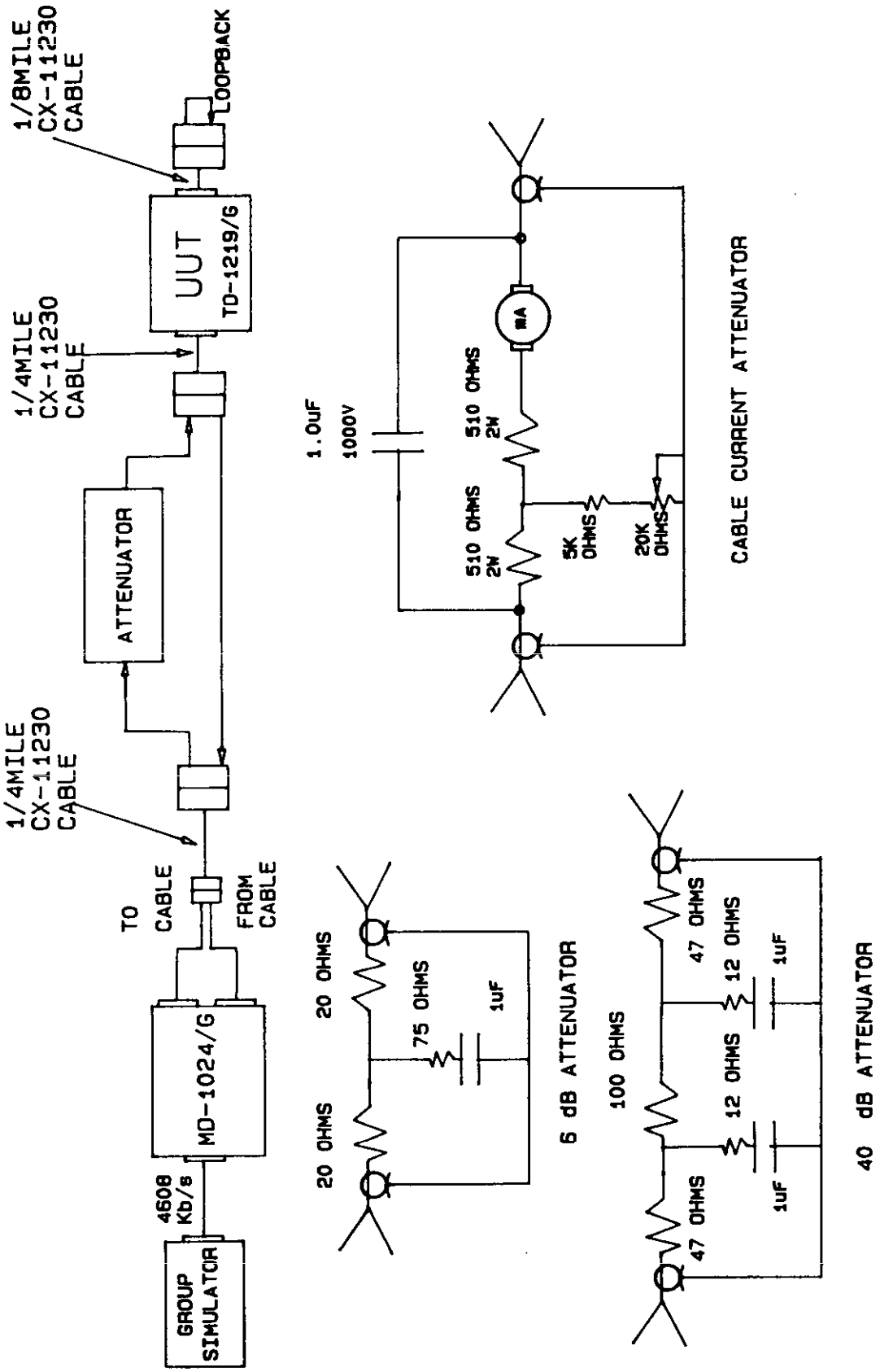


FIGURE 6. Receiver sensitivity; power supply variation; fault detector sensitivity.

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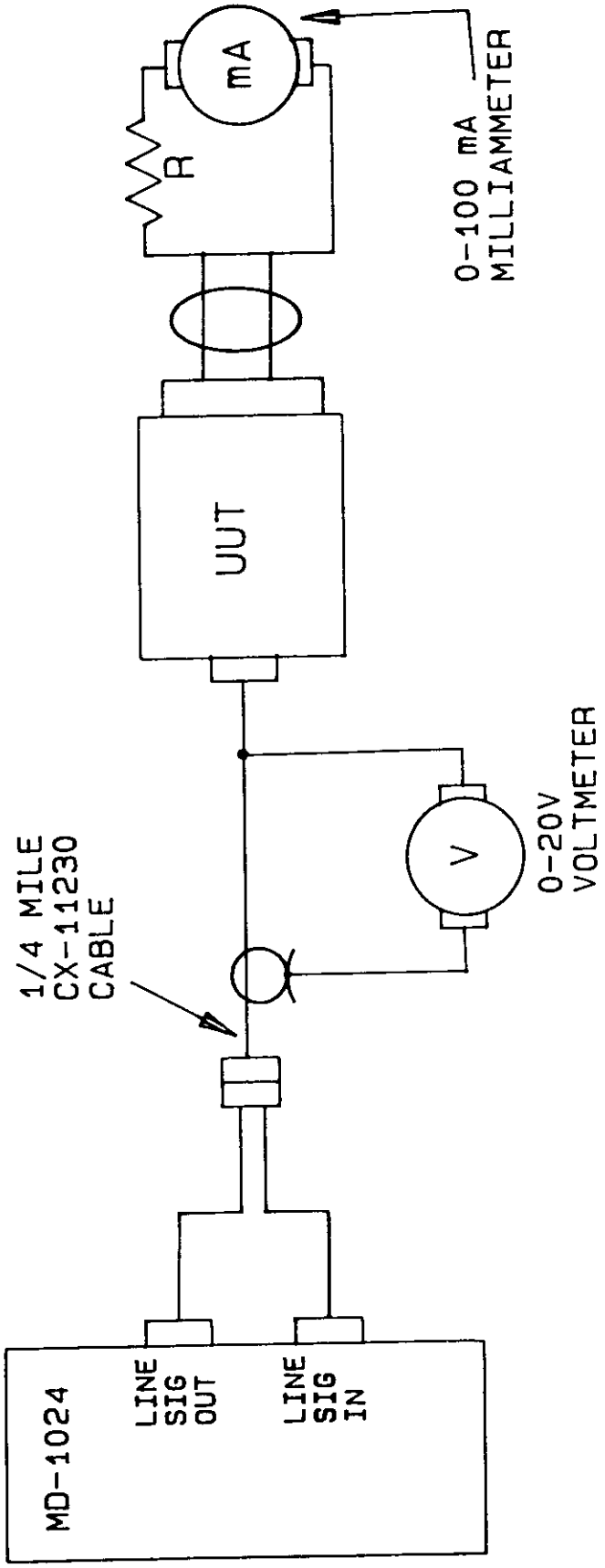


FIGURE 7. Fault-detector sensitivity and resistance.

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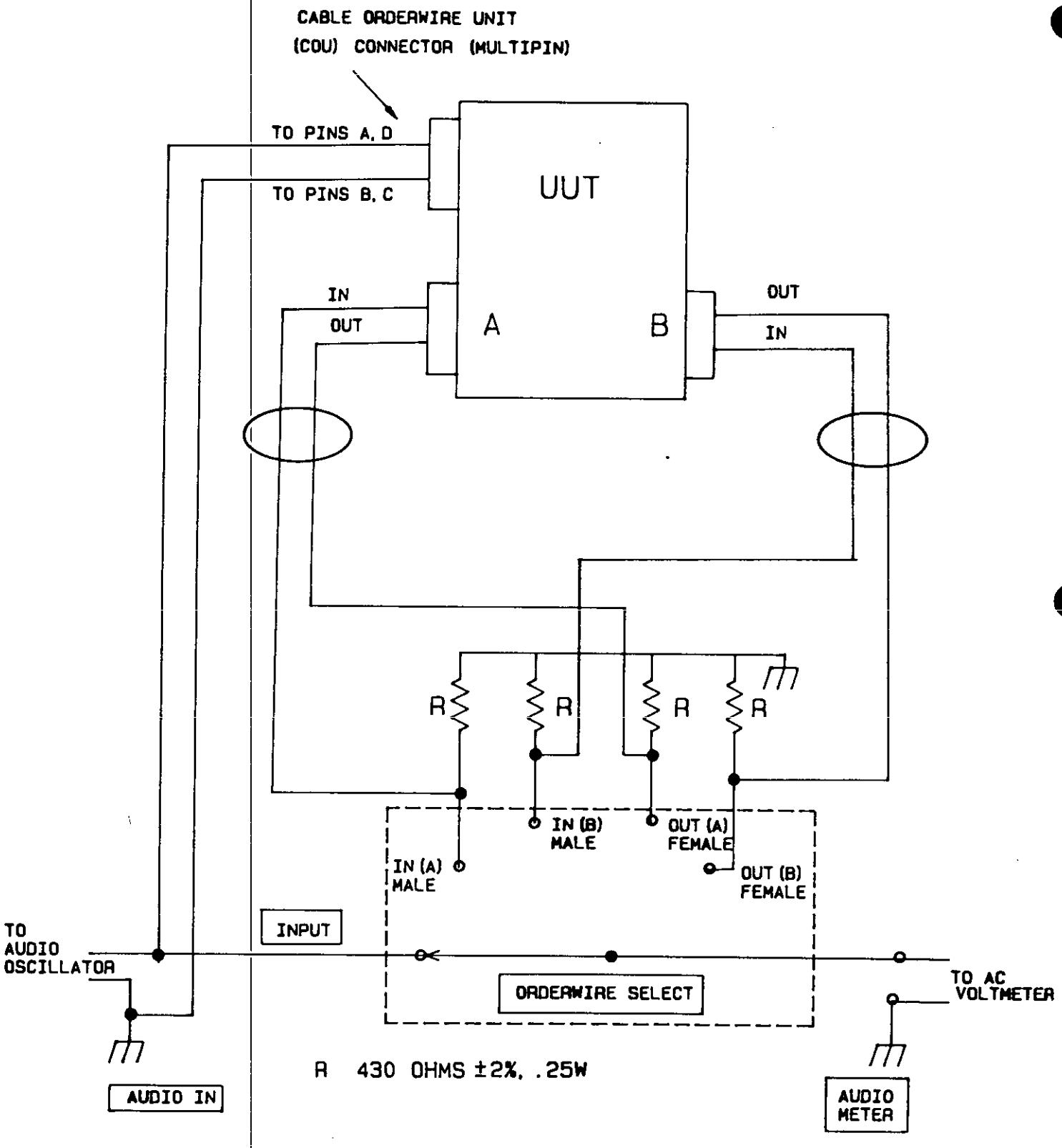


FIGURE 8. Orderwire connector.

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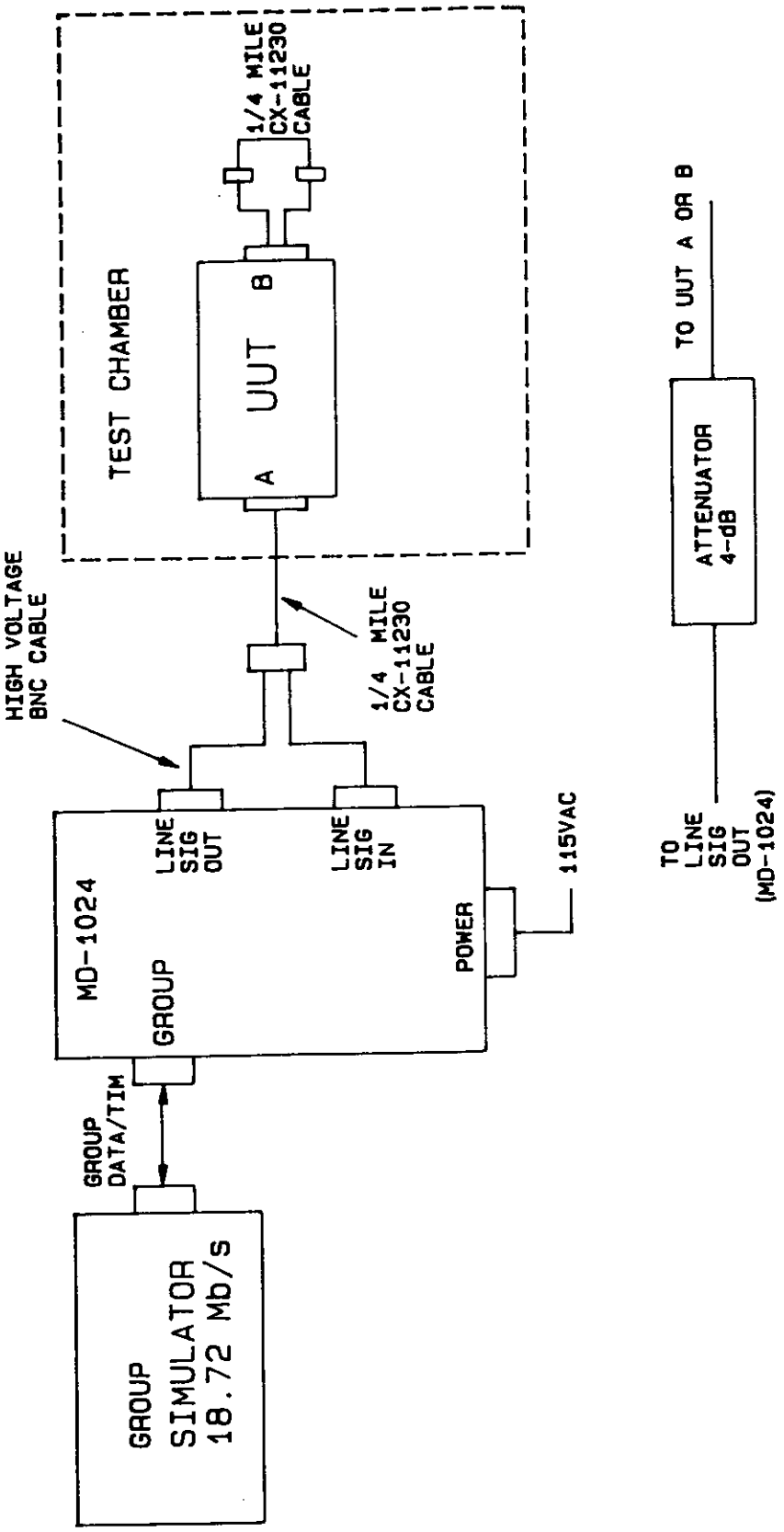


FIGURE 9. Environmental signal detection.

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